

The Khronon Mass Scale from Recombination: Two Numerical Relations Connecting MOND, CMB, and Dark Matter

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In the Blanchet–Skordis Khronon dark matter theory, the inverse mass parameter $\mu^{-1} = 22.3 \text{ Mpc}$ is fitted to galaxy rotation curves and CMB data, but its theoretical origin is unknown. We report two numerical relations, apparently absent from the literature, that express μ in terms of fundamental cosmological quantities: (i) $\mu c^2 = a_0 (1 + z_{\text{dec}})$, relating the Khronon mass to the MOND acceleration blueshifted to photon decoupling ($\mu^{-1} = 22.25 \text{ Mpc}$, 0.2% agreement); and (ii) $\mu = 2\pi(1 + \Omega_b)/r_d$, relating the Khronon Compton wavelength to the baryon-loaded sound horizon ($\mu^{-1} = 22.31 \text{ Mpc}$, 0.04% agreement). The two relations are structurally independent yet mutually consistent to 0.27%. Both connect a galactic-scale parameter (μ or a_0) to recombination-epoch quantities (z_{dec} , r_d , Ω_b), suggesting that the Khronon mass is not a free parameter but is fixed by the physics of baryon–photon decoupling.

INTRODUCTION

The dark matter problem admits two complementary phenomenological descriptions. On galactic scales, the MOND acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ ms}^{-2}$ governs the baryonic Tully–Fisher relation and the radial acceleration relation [1, 2]. On cosmological scales, the Λ CDM paradigm reproduces the CMB power spectrum and large-scale structure with a pressureless cold dark matter component, $\Omega_{\text{DM}} h^2 = 0.120$ [4]. Bridging these regimes has proven difficult: theories that reproduce MOND phenomenology generically fail CMB constraints, and vice versa.

A notable exception is the Blanchet–Skordis (BS) Khronon theory [6, 7], which extends the Skordis–Złośnik action [5] with a Dirac–Born–Infeld (DBI) kinetic structure. The theory contains two sectors: a galactic sector $J(Y)$ that recovers MOND, and a cosmological sector $K(Q) = \mu^2(Q - 1)^2$ that provides CDM-like perturbations through ghost condensation. The single parameter μ controls the transition from the modified-gravity regime to the Newtonian regime; its inverse $\mu^{-1} = 22.3 \text{ Mpc}$ is determined by fitting to CMB and rotation curve data [7].

Despite the theory’s empirical success, the origin of μ remains unexplained. Dimensional analysis suggests $\mu \sim H_0/c$, giving $\mu^{-1} \sim c/H_0 \approx 4450 \text{ Mpc}$ —a factor of ~ 200 too large, and incompatible with CMB data [7]. In this Letter, we report two numerical relations that pin down μ in terms of well-measured cosmological quantities with sub-percent accuracy.

RELATION I: MOND ACCELERATION AT DECOUPLING

We propose

$$\mu c^2 = a_0 (1 + z_{\text{dec}}), \quad (1)$$

where $z_{\text{dec}} = 1089.92$ is the photon decoupling redshift [4]. This can be read as: *the Khronon mass scale equals the MOND acceleration blueshifted to the epoch of last scattering, expressed in natural units.*

Inverting Eq. (1) gives the Khronon Compton wavelength,

$$\mu^{-1} = \frac{c^2}{a_0 (1 + z_{\text{dec}})} = \frac{(2.998 \times 10^8)^2}{1.2 \times 10^{-10} \times 1090.92} = 22.25 \text{ Mpc}. \quad (2)$$

Compared to the BS2025 fitted value $\mu_{\text{fit}}^{-1} = 22.3 \text{ Mpc}$ [7], the agreement is

$$\frac{|\mu^{-1} - \mu_{\text{fit}}^{-1}|}{\mu_{\text{fit}}^{-1}} = 0.2\%. \quad (3)$$

Physical interpretation.—The quantity $a_0(1 + z_{\text{dec}})$ is the comoving MOND acceleration at decoupling. In MOND phenomenology, a_0 marks the transition between Newtonian and deep-MOND dynamics at $z = 0$; at redshift z , this transition scale blueshifts to $a_0(1 + z)$ [9]. Relation (1) states that the Khronon mass μ is set by this transition scale evaluated precisely at the epoch when baryons decouple from photons. This is physically natural: μ controls the MOND–Newton transition in the Khronon theory, and its value should be determined by the cosmological epoch at which the relevant degrees of freedom (baryonic perturbations) become dynamically independent.

RELATION II: BARYON-LOADED SOUND HORIZON

Independently, we find

$$\mu = \frac{2\pi (1 + \Omega_b)}{r_d}, \quad (4)$$

where $r_d = 147.09 \text{ Mpc}$ is the comoving sound horizon at the baryon drag epoch [4] and $\Omega_b = 0.0493$ is the baryon

density parameter. Inverting,

$$\mu^{-1} = \frac{r_d}{2\pi(1+\Omega_b)} = \frac{147.09}{2\pi \times 1.0493} = 22.31 \text{ Mpc}. \quad (5)$$

The agreement with the fitted value is

$$\frac{|\mu^{-1} - \mu_{\text{fit}}^{-1}|}{\mu_{\text{fit}}^{-1}} = 0.04\%. \quad (6)$$

Physical interpretation.—The sound horizon r_d is the comoving distance traveled by acoustic waves in the baryon–photon plasma from the Big Bang to recombination. Relation (4) states that the Khronon Compton wavelength is $r_d/2\pi$ corrected by the baryon loading factor $(1+\Omega_b)$. In other words, μ^{-1} is the *reduced acoustic wavelength* of the sound horizon, adjusted for the gravitational drag of baryons on the plasma oscillations. This connects the Khronon mass—which governs galactic-scale MOND phenomenology—directly to the BAO scale imprinted at recombination.

MUTUAL CONSISTENCY

Relations (1) and (4) are structurally independent: the first involves a_0 and z_{dec} ; the second involves r_d and Ω_b . Neither contains the other’s characteristic quantities. Yet the two predictions agree to

$$\frac{|22.25 - 22.31|}{22.31} = 0.27\%, \quad (7)$$

well within the observational uncertainty on a_0 (typically 5–10% [2, 3]) and on r_d (0.2% [4]).

Equating the two relations yields a *consistency condition* linking MOND, CMB, and baryon physics:

$$a_0 = \frac{2\pi(1+\Omega_b)c^2}{r_d(1+z_{\text{dec}})}. \quad (8)$$

Evaluating with Planck 2018 parameters gives

$$a_0^{\text{pred}} = \frac{2\pi \times 1.0493 \times (2.998 \times 10^8)^2}{4.536 \times 10^{24} \times 1090.92} = 1.197 \times 10^{-10} \text{ m s}^{-2}, \quad (9)$$

where $r_d = 147.09 \text{ Mpc} = 4.539 \times 10^{24} \text{ m}$. This agrees with the empirical MOND value $a_0 = (1.20 \pm 0.02_{\text{stat}} \pm 0.24_{\text{sys}}) \times 10^{-10} \text{ m s}^{-2}$ [2, 3] to 0.3%.

Equation (8) is a parameter-free prediction: *if* the Khronon mass is set by recombination physics, *then* the MOND acceleration is calculable from r_d , z_{dec} , Ω_b , and c alone. This provides a sharp test—any future refinement of a_0 , r_d , or Ω_b must preserve Eq. (8).

THE FACTOR-OF-200 HIERARCHY

As noted in Sec. , the naive dimensional estimate $\mu \sim H_0/c$ overshoots by a factor of ~ 200 . Relation (1) makes

this hierarchy transparent:

$$\frac{\mu}{H_0/c} = \frac{a_0(1+z_{\text{dec}})}{cH_0} = \frac{(1+z_{\text{dec}})}{2\pi} \times \frac{a_0}{a_0^{\text{M}}}, \quad (10)$$

where $a_0^{\text{M}} \equiv cH_0/(2\pi) \approx 1.04 \times 10^{-10} \text{ m s}^{-2}$ is Milgrom’s cosmological coincidence value [1, 9]. Since $a_0/a_0^{\text{M}} \approx 1.15$ empirically,

$$\frac{\mu}{H_0/c} \approx \frac{1091}{2\pi} \times 1.15 \approx 200. \quad (11)$$

The factor of 200 is thus not a mysterious large number but the product of two well-understood ingredients: $(1+z_{\text{dec}})/(2\pi)$, a purely geometrical–cosmological ratio, and the $O(1)$ correction a_0/a_0^{M} .

This also clarifies why $\mu_0 = H_0/c$ —the value motivated by dimensional analysis and used in Refs. [10, 11]—is excluded by CMB data: the physics of recombination inserts a factor of $(1+z_{\text{dec}})/(2\pi)$ that cannot be neglected.

DISCUSSION

Two routes to one number.—The two relations reported here arrive at $\mu^{-1} \approx 22.3 \text{ Mpc}$ via completely different physical ingredients. Relation I uses the MOND acceleration a_0 —a galactic observable—and the decoupling redshift. Relation II uses the sound horizon r_d —a cosmological observable—and the baryon fraction. Their mutual consistency (Eq. (8)) is a nontrivial constraint that any microscopic derivation of μ must satisfy.

Implications for the BS theory.—If either relation holds as an identity rather than a coincidence, it reduces the free parameter count of the BS Khronon theory from two (a_0, μ) to one (a_0 alone, with μ derived). Furthermore, if the consistency condition (8) is exact, a_0 itself is determined by cosmological parameters, leaving the theory with *zero* free parameters in the dark sector beyond ΛCDM inputs.

Connection to Milgrom’s FUNDAMOND program.—Milgrom has long advocated that $a_0 \sim cH_0$ reflects a deep connection between local and cosmological dynamics [8, 9]. Equation (8) refines this conjecture: the connection runs not through H_0 directly but through the sound horizon r_d , the baryon fraction Ω_b , and the decoupling redshift z_{dec} . All three are determined by the baryon–photon plasma physics at recombination, suggesting that the MOND scale a_0 is an *imprint* of recombination rather than a reflection of the present-day Hubble rate.

What remains open.—We have not derived either relation from first principles. A microscopic derivation would need to show that ghost condensation in the Khronon sector locks onto the acoustic scale at recombination. One plausible mechanism is that the Khronon field, being a

timelike scalar coupled to the metric, develops its condensate precisely when the baryon–photon plasma decouples and the sound speed drops sharply. The coincidence $\mu^{-1} \sim r_d/(2\pi)$ suggests that the Khronon “freezes in” at a wavelength corresponding to the last acoustic oscillation, analogous to how axion dark matter freezes at the QCD scale. We leave a detailed derivation to future work.

Falsifiability.—Both relations make precise, testable predictions. Relation I predicts $\mu^{-1} = c^2/[a_0(1 + z_{\text{dec}})]$; any independent measurement of μ from CMB or large-scale structure data must match. The consistency condition (8) predicts $a_0 = 1.197 \times 10^{-10} \text{ m s}^{-2}$; this is falsifiable by future high-precision measurements of the radial acceleration relation (e.g., from DESI or Euclid weak-lensing samples).

SUMMARY

We have reported two numerical relations:

$$(I) \quad \mu c^2 = a_0 (1 + z_{\text{dec}}), \quad (12)$$

$$(II) \quad \mu = 2\pi (1 + \Omega_b)/r_d, \quad (13)$$

each reproducing the BS-fitted Khronon mass $\mu^{-1} = 22.3 \text{ Mpc}$ to better than 0.3%. Their mutual consistency yields a parameter-free prediction for the MOND acceleration, $a_0 = 2\pi(1 + \Omega_b) c^2/[r_d(1 + z_{\text{dec}})] = 1.20 \times 10^{-10} \text{ m s}^{-2}$. If confirmed as exact relations, they eliminate the last free parameter of the Khronon dark matter theory and anchor the MOND scale to the physics of baryon–photon decoupling.

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